MATH470 REPORT

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1. INTRODUCTION

Copulas are a key point of interest in understanding the scale-free dependence structure of multivariate distributions, by joining - or "coupling" - their marginal distribution functions with their joint distribution function in such a way that their one dimensional marginal distributions are uniform on the interval $[0, 1]$. In this paper we focus our attention to the bivariate case of the joint distribution of the largest and second largest order statistics. Motivated by a data driven example of rainfall in Aix, France, we calculate the copula of the largest and second largest order statistics of this data, assuming a GEV fit to the data. We find that this copula is in fact independent of any parameters of the GEV distributions, and then prove that for an independent, identically distributed sample of size n from any distribution, this is the limit copula of $(X_{(n)}, X_{(n-1)})$ as $n \to \infty$.

1.1. Motivation.

In September 1992, a monsoon storm swept the south of France setting a record high death toll of over 32 people and generating $1 million in damage and emergency funds [1],[8]. Aix-en-Provence in particular measured a record high 222mm of rain during this storm [3]. Extreme weather events like this can feel impossible to predict when they don't follow the typical weather patterns of the region. Extreme Value Theory is a crucial tool for prediction and prevention of such floods or other extreme weather phenomenon (see [4], [5] for more background on extreme value theory).

1.2. GEV background.

Let $X_{(n)} = \max\{X_1, \ldots, X_n\}$ where $X_1, \ldots, X_n$ is a sequence of independent random variables from some distribution, F. We call n the *block size*, and $X_{(n)}$ the corresponding *block maximum*. Then, for suitable sequences of constants a_n and b_n and large enough n, $P((X_{(n)} - b_n)/a_n \leq z) \approx G(z)$, where G is a member of the *Generalized Extreme Value* (GEV) family:

$$G(z) = \exp\{-[1 + \xi(\frac{z-\mu}{\sigma})]^{-1/\xi}\}, \tag{1}$$

defined on $z : 1 + \xi(z - \mu)/\sigma > 0$, where $\mu, \xi \in \mathbb{R}$ and $\sigma > 0$ (Theorem 3.1.1 [4]). Despite the normalizing constants often being unknown in practice, $X_{(n)}$ itself can be estimated by a different member $G^*(z)$ of the same family. Since the parameters must be estimated, the differences between the distributions of $X_{(n)}$ and its scaled counterpart are irrelevant in practice (p. 49 [4]).

A quantity of interest often associated with the GEV distribution is the *return level*. The *k-return level* for a block of size n is the value which is expected to be exceeded once every k blocks. In other words, it is the value which is exceeded by the block maximum in any

particular block with probability $1/k$. The return level is given by the following formula:

$$r = \begin{cases} \mu - \frac{\sigma}{\xi}[1 - y_k^{-\xi}] & \xi \neq 0, \\ \mu - \sigma \ln(y_k) & \xi = 0, \end{cases} \tag{2}$$

where $y_k = -\ln(1 - \frac{1}{k})$ (p. 49 [4]). The return level can be readily estimated by simply plugging in the maximum likelihood estimates for the parameters $\hat{\mu}, \hat{\sigma}, \hat{\xi}$.

In the context of weather phenomenon, return level can be used to predict when rainfall may exceed a dangerous amount, potentially causing a flood, such as in 1992.

2. Introducing our data

In this paper we explore daily rainfall data collected from 180 different weather stations in France measured from 1976 to 2015 (provided by Météo France [3]). In particular, we focus on station number 18, stationed in Aix-en-Provence, France.

If we assume that rainfall on any given day is an independent event, then a Generalized Extreme Value (GEV) distribution is an appropriate model to fit to both the yearly and monthly block maxima of this data.

The choice of block size is critical when making inferences from a GEV model. For example, with 40 years of data, a block size of one year yields 40 block maxima, where as choosing monthly maxima would give us a block maxima sample of size 480. Since model inferences, which are based on the asymptotic normality of our estimators, come from the limit behavior of the size of our sample, a smaller block size will provide less *variation* in our parameter estimates, due to the larger sample size. However, there is a trade off: Since the limit distribution of $M_n = \max\{X_1, \ldots, X_n\}$ is as the block size $n \to \infty$, a smaller block size, e.g. monthly maxima instead of yearly, means the approximation in the limit may be poor, yielding greater *bias* in parameter estimates.

Using the ismev package in R, we can compare the maximum likelihood estimates of the GEV parameters, μ, σ, ξ, of both a yearly and monthly block size:

MLE	Yearly blocks	Monthly blocks
$\hat{\mu}$	47.67	11.71
$\hat{\sigma}$	17.13	10.91
$\hat{\xi}$	0.215	0.271

Standard Error	Yearly blocks	Monthly blocks
$\hat{\mu}$	3.06	0.60
$\hat{\sigma}$	2.45	0.50
$\hat{\xi}$	0.127	0.050

Observe how the standard errors of our parameter estimates for the monthly blocks are significantly lower than those for the yearly blocks. However, as shown in the qqplots below, the deviation in figure 2 may suggest that a GEV model is a better fit for yearly block size. These models can now be used to estimate return levels. In light of the variance-bias trade-off, one can use profile likelihood estimation for greater accuracy in confidence intervals for estimates of values such as return level (p. 60 [4]).

The tail behavior of profile log-likelihood (figure 3) takes into account the decreasing accuracy as we try to predict a larger return period, since it becomes increasingly harder to predict behavior based on only 40 years of data.

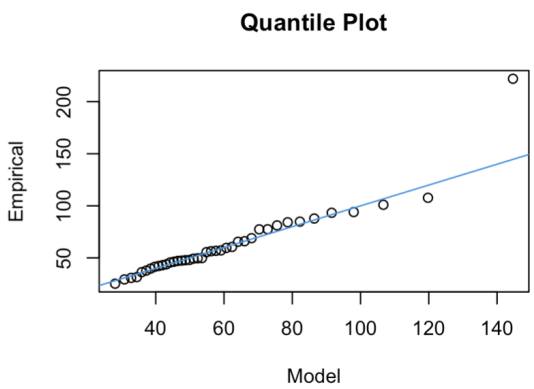

FIGURE 1. Quantile plot for yearly block maxima

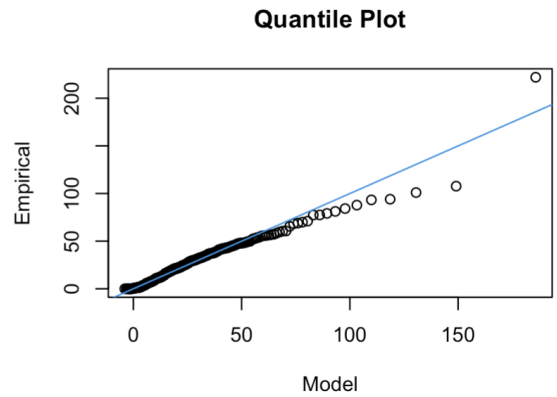

FIGURE 2. Quantile plot for monthly block maxima

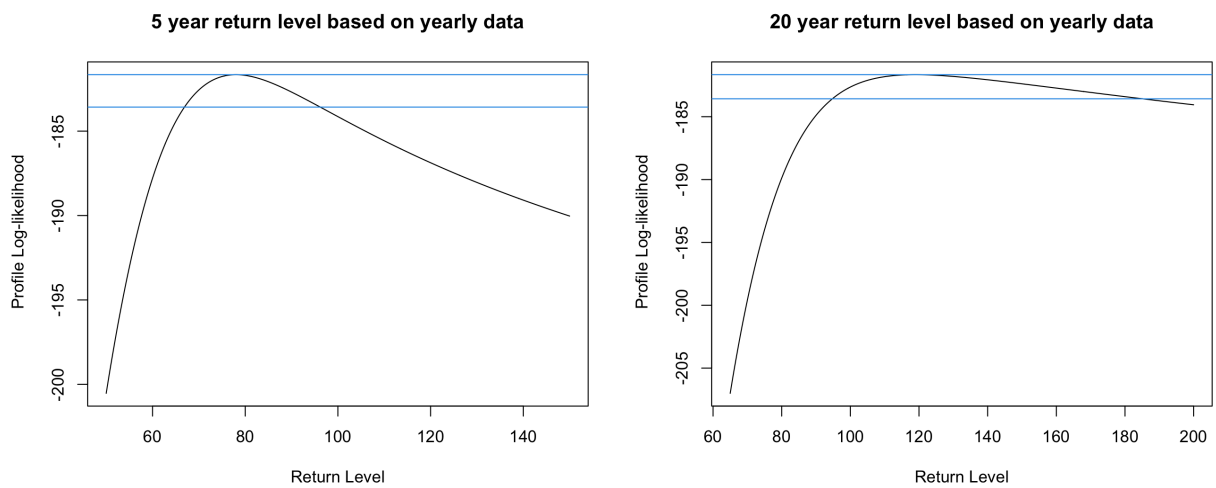

FIGURE 3. Profile log-likelihood estimate for 5 and 50 year return levels on yearly block maxima.

In order to try to make better inferences from our data, we can double our sample size while still maintaining a yearly block size by looking at both the largest and second largest yearly block maxima. Doing so introduces many interesting dependencies which may be present between these two variables.

3. Second largest order statistics

One may also be concerned with how the largest and second largest rainfall measurements are related to each other.

Let $X_{(n)} = \max(X_1, \ldots, X_n)$ be such that the limiting distribution of $X_{(n)}$ (suitably re-scaled) is GEV with parameters ξ, μ, σ as $n \to \infty$ (as in equation 1). Then Theorem 3.4 from [4] gives us that the second largest order statistic $X_{(n-1)}$, after suitable re-scaling, also

follows a GEV distribution as $n \to \infty$. We can similarly absorb the scaling constants so that the limit distribution of $X_{(n-1)}$ is approximately GEV itself.

Using a Poisson limit process we find the joint CDF of $(X, Y) := (X_{(n)}, X_{(n-1)})$ to be:

$$F(x,y) = \begin{cases} e^{-\tau(x)} & y \geq x \\ e^{-\tau(y)}[1 + \tau(y) - \tau(x)] & y < x, \end{cases} \tag{3}$$

where $\tau(x) = [1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi}$ (see A.1 for calculation details and sections 3.5 and 7 in [4] for more background on the Poisson limit process).

Remark 3.1. Note that τ is defined when $1 + \xi(\frac{x-\mu}{\sigma}) > 0$. If $\xi > 0$ then we have a lower bound on the support: $x > \mu - \sigma/\xi$, and if $\xi < 0$ then we have an upper bound of $x > \mu - \sigma/\xi$, so $\tau(x) = 0$ for $x \geq \mu - \sigma/\mu$.

Their marginal distributions can be computed as follows:

$$\begin{aligned} F_X(x) &= P(X_{(n)} \leq x, X_{(n-1} \leq \infty) = F(x, \infty) = e^{-\tau(x)} \\ F_Y(y) &= F(\infty, y) \\ &= \lim_{x\to\infty} e^{-\tau(y)}(1 + \tau(y) - \tau(x)) \\ &= e^{-\tau(y)}[1 + \tau(y) - \lim_{x\to\infty} \tau(x)] \\ &= e^{-\tau(y)}[1 + \tau(y)] \end{aligned} \tag{4}$$

as $\lim_{x\to\infty} \tau(x) = 0$ if $\xi > 0$. If $\xi < 0$, then by remark 3.1, $\tau(x) = 0$ for large enough x.

To explore the dependence structure, we'd like to introduce a copula model for the largest and second largest order statistics.

Definition 3.2. Let $(X, Y) \sim F(x, y)$ with marginal distributions F_X, F_Y, respectively. Then the *copula* of (X, Y) is defined as the distribution of $(U, V) := (F_X(X), F_Y(Y))$,

$$C(u, v) = P(U \leq u, V \leq v),$$

Note that $C : [0, 1]^2 \to [0, 1]$ as U, V are both uniform on $[0, 1]$.

By using Sklar's Theorem, the copula of (X, Y) can readily be recovered as

$$C(u, v) = F(F_X^{-1}(u), F_Y^{-1}(v)).$$

If $(U, V) \sim C(u, v)$ where $C(u, v)$ is a copula model, then $F_u(v) := \frac{\partial}{\partial u} C(u, v)$ is the conditional probability $P(V \leq v | U = u)$. This can be used to draw a theoretical sample from C. For a more detailed review of copulas, see [7]. When marginal distributions are not available we make use of the following estimation:

Definition 3.3. Let (X_i, Y_i), $i = 1, \ldots, n$ be a sample from (X, Y). Then the *empirical copula* of (X, Y) is

$$C_n(u, v) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(\frac{R_i}{n+1} \leq u, \frac{S_i}{n+1} \leq v),$$

where R_i, S_i are the ranks of X_i and Y_i respectively.

3.4. **Empirical copula observations from the data.** For station number 18, stationed in Aix-en-Provence, France, we can extract both the largest and second largest monthly rainfall measurements, giving us a sample (x_i, y_i) where x_i is the largest rainfall measurement and y_i is the second largest, for months $i = 1, \ldots, 480$. There are then two natural ways to transform our data to make observations about the copula structure between $X_{(n)}, X_{(n-1)}$:

(1) If we impose a GEV fit to the data, we can estimate $\hat{F_X}$ and $\hat{F_Y}$ by substituting the maximum likelihood estimates $\hat{\xi}, \hat{\sigma}, \hat{\mu}$ obtained in section 2 into the marginal distributions given by equation 4. This gives us a *model based copula sample*: $\hat{F}_X(x_i), \hat{F}_Y(y_i)$. Note that this method assumes a fit to our data which may or may not be true.
(2) To avoid fitting a model to our data which may or may not be true, we can also estimate the marginal distributions of $X_{(n)}$ and $X_{(n-1)}$ non-parametrically by using their empirical CDFs[1] to obtain the *empirical copula sample*: $(\frac{R_{x_i}}{n+1}, \frac{R_{y_i}}{n+1})$, where R_{x_i}, R_{y_i} are the ranks of x_i, y_i respectively.

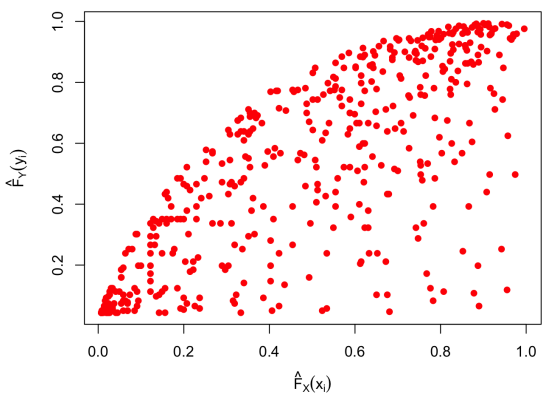

FIGURE 4. Model Based Copula from Monthly Block Maxima

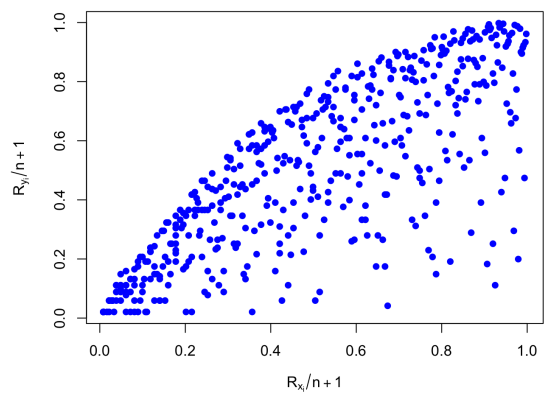

FIGURE 5. Empirical Copula from Monthly Block Maxima

Observation 3.5. Both the empirical copula sample and the model based copula sample follow the same pattern where the mass is distributed beneath a specific curve. This suggests that the underlying copula of this distribution is governed by some curve.

This leads to the two following natural questions:

(1) Can we find this curve?
(2) Is this curve (and the underlying copula) independent of any parameters (namely, μ, σ, ξ) from the GEV model which we fit to our data?

4. Calculating the curve

Theorem 4.1. Let $(X_{(n)}, X_{(n-1)})$ be the largest and second largest order statistics of an i.i.d. sample $X_1, \ldots, X_n$ such that the limit distribution of $(X_{(n)}, X_{(n-1)})$ is GEV with parameters

[1]The empirical CDF for a sample of size n is defined as $F_n(x) = \frac{1}{n+1} \sum_{i=1}^{n} \mathbb{1}(x_i \leq x)$.

ξ, μ, σ as described in equation 3, as $n \to \infty$. Then the copula $C(u,v) = F(F_X^{-1}(u), F_Y^{-1}(v))$ is independent of the parameters μ, σ, ξ.

Proof. It is already known that the copula of $(X_{(n)}, X_{(n-1)})$ is independent of μ and σ because copulas are invariant to strictly increasing transformations (Theorem 2.4.3 [7]). Now I claim that the copula is further independent of ξ as well.

Let $(U,V) = F_X(X_{(n)}), F_Y(X_{(n-1)})$ so that $C(u,v)$ is the joint distribution of (U,V).

By using equation 3, observe that

$$\begin{aligned} C(u,v) &= F(F_X^{-1}(u), F_Y^{-1}(v)) \\ &= \begin{cases} e^{-\tau(F_X^{-1}(u))} & F_Y^{-1}(v) \geq F_X^{-1}(u) \\ e^{-\tau(F_Y^{-1}(v))}(1 + \tau(F_Y^{-1}(v)) - \tau(F_X^{-1}(u))) & F_Y^{-1}(v) < F_X^{-1}(u) \end{cases} \end{aligned}$$

Note that even though $F_Y(y)$ is not explicitly invertible, it *is* monotonically increasing so the notion of an inverse makes numerical sense.

One can then show that

$$\frac{\partial}{\partial u}[e^{-\tau(F_X^{-1}(u)}] = 1$$

$$\frac{\partial}{\partial u}[e^{-\tau(F_Y^{-1}(v))}(1 + \tau(F_Y^{-1}(v)) - \tau(F_X^{-1}(u)))] = \frac{1}{u}e^{-\tau(F_Y^{-1}(v))}$$

Since F_Y, F_X are both monotonic, we have $F_Y^{-1}(v) \geq F_X^{-1}(u) \iff v \geq u(1 - \ln u)$, hence:

$$\frac{\partial}{\partial u}C(u,v) = \begin{cases} 1 & v \geq u(1 - \ln u) \\ \frac{1}{u}e^{-\tau(F_Y^{-1}(v))} & v < u(1 - \ln u) \end{cases}$$

The curious reader can see the appendix A.2 for the calculation details.

Let $F_u(v) = \frac{\partial}{\partial u}C(u,v) = P(V \leq v | U = u)$. This is the conditional probability of V given U. Recall, we can always write $V = F_U^{-1}(W)$ where $W \sim U[0,1]$ is independent. Another series of calculations (see A.3) shows that $F_u^{-1}(w) = uw(1 - \ln uw)$.

So for an independent random variable $W \sim U[0,1]$, we have that

$$\begin{aligned} C(u,v) &= P(U \leq u, V \leq v) \\ &= P(U \leq u, UW(1 - \ln UW) \leq v) \\ &= \int_0^u P(UW(1 - \ln UW) \leq v | U = t)dt \end{aligned}$$

Now let $\Psi_t : (0,1) \to \mathbb{R}$ be defined as $\Psi_t(w) = tw(1 - \ln(tw))$ for some $t \in (0,1)$. We restrict the domain to $(0,1)$ because $W \sim U[0,1]$ so $0 < W < 1$.

Note that $\frac{d}{dw}\Psi_t(w) = -t\ln(tw) > 0$ for $0 < w < \frac{1}{t} < 1$ whenever $t \in (0,1)$, hence $\Psi_t(w)$ is strictly increasing, and continuous, for $w \in (0,1)$. So even though $\Psi_t(w)$ is not explicitly invertible, one can numerically invert it on $(0,1)$.

Now we compute the integrand by taking the probability with respect to W and fixing $t \in (0,1)$:

$$\begin{aligned} P(UW(1-\ln(UW) \leq v | U = t) &= P_W(tW(1-\ln tW) \leq v) \\ &= P_W(\Psi_t(W) \leq v) \\ &= P_W(W \leq \Psi_t^{-1}(v)) \\ &= \begin{cases} \Psi_t^{-1}(v) & v < t(1-\ln t) \\ 1 & v \geq t(1-\ln t) \end{cases} \end{aligned}$$

Plugging this back into our integral we find that

$$\begin{aligned} C(u,v) &= \int_0^u P(UW(1-\ln(UW)) \leq v | U = t) \\ &= \int_0^u \mathbb{1}\{v \geq t(1-\ln(t))\} + \mathbb{1}\{v < t(1-\ln(t))\}\Psi_t^{-1}(v)dt, \end{aligned} \tag{5}$$

which is entirely independent of any parameters from the GEV distribution.

□

We can now easily generate a sample (u_i, v_i) from this theoretical distribution by letting u_i be uniform on the interval $[0,1]$ and setting $v_i = F_{u_i}^{-1}(w_i)$ for independent uniform samples w_i on $[0,1]$.

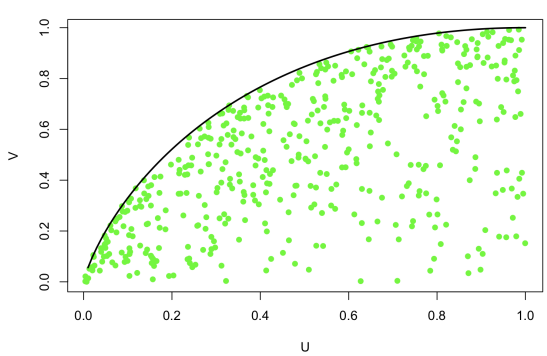

FIGURE 6. Theoretical Copula Generated from $C(u,v)$

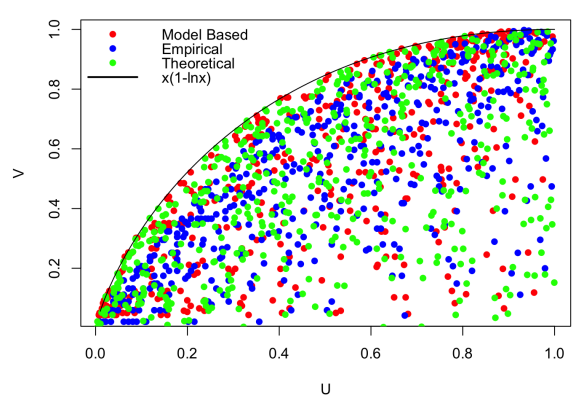

FIGURE 7. Comparing all 3 Copula Samples

Observe that the points sampled from this copula model all lie below the curve

$$f(x) = x(1-\ln(x)).$$

This is because for all i we have,

$$\begin{aligned} (u_i, v_i) &\in (0,1) \times (F_{u_i}^{-1}(0), F_{u_i}^{-1}(1)) \\ &= (0,1) \times (0, u_i(1-\ln u_i)) \end{aligned}$$

It is also interesting to note that both the model based and empirical copula of our actual data points for the largest and second largest rainfall measurements follow a similar pattern.

5. This is the limit curve!

As shown in figure 7, the mass of the theoretical copula of $(X_{(n)}, X_{(n-1)})$, when $(X_{(n)}, X_{(n-1)})$ is GEV in the limit, appears to lie below the curve $f(x) = x(1 - \ln(x))$. We'd like to investigate whether this curve is the result of limit behavior as the sample size n goes to infinity.

Theorem 4.1 also tells us that the copula of $(X_{(n)}, X_{(n-1)})$ is independent of any GEV parameters, suggesting that this curve may govern the dependence structure of the largest and second largest order statistics from any distribution, not just one which is GEV in the limit.

For $i = 1, \ldots, n$, let $X_i \sim F$ and $U_i \sim U[0, 1]$ be independent samples. Then the copula of $(X_{(n)}, X_{(n-1)})$ is the same as that of $(U_{(n)}, U_{(n-1)})$. This is due to the fact that both copulas and the rank of X_i are invariant under strictly increasing transformations.

Hence it suffices to simply consider an independent sample of uniform random variables.

5.1. **Simulation.** We can run a simulation in R of n independent $U[0, 1]$ sample points, and plot the empirical copula of $(U_{(n)}, U_{(n-1)})$. Figure 8 shows the empirical copula behavior for two samples with $n = 2$ versus $n = 100$. The points appear to approach the curve $f(x) = x(1 - \ln x)$ as n grows, providing evidence of this hypothesis.

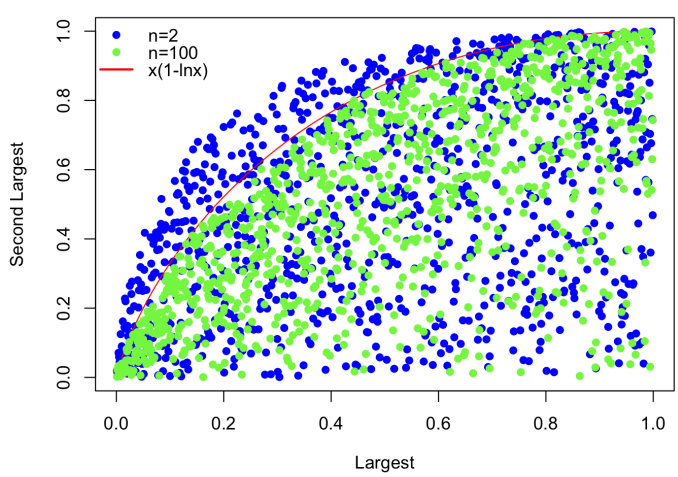

Figure 8. Empirical copulas of $(U_{(n)}, U_{(n-1)})$ for two simulated samples of size 2 and 100 from $U[0, 1]$.

5.2. **Return to theory.** Let $U_1, \ldots, U_n \sim U[0,1]$ be independent. Then from Theorem 5.4.6 of [2], the joint probability density of $(U_{(n)}, U_{(n-1)})$ is

$$f(x,y) = n(n-1)y^{n-2}\mathbb{1}(0 < y < x < 1).$$

Integrating this we find that the joint distribution is

$$\begin{aligned} F(x,y) &= \int_0^y \int_0^x n(n-1)s^{n-2}\mathbb{1}(0 < s < t < 1)dtds \\ &= \begin{cases} 0 & x \text{ or } y < 0 \\ nxy^{n-1} - (n-1)y^n & 0 < y \leq x \\ x^n & 0 < x \leq y < 1 \\ 1 & x \text{ and } y \geq 1. \end{cases} \end{aligned}$$

From this we can extract the marginal distributions:

$$\begin{aligned} F_1(x) &= P(U_{(n)} \leq x) = F(x,1) = x^n \\ F_2(y) &= P(U_{(n-1)} \leq y) = F(1,y) = ny^{n-1} - (n-1)y^n \end{aligned}$$

Then the copula of $(U_{(n)}, U_{(n-1)})$ is $C(u,v) = F(F_1^{-1}(u), F_2^{-1}(v))$. The inverse of F_1 is easy to compute, and while $F_2(y)$ is not explicitly invertible, it is increasing on $(0,1)$, so a numerical inverse makes sense.

Let $\phi(x) = F_2(F_1^{-1}(x)) = nx^{(n-1)/n} - (n-1)x$.

Then since F_1, F_2 are monotonically increasing on $(0,1)$, we have $F_2^{-1}(v) \leq F_1^{-1}(u) \iff v \leq \phi(u)$, so we can calculate:

$$C(u,v) = \begin{cases} 0 & u \text{ or } v < 0 \\ nu^{1/n}[F_2^{-1}(v)]^{n-1} - (n-1)[F_2^{-1}(v)]^n & 0 < v \leq \phi(u) \\ u & \phi(u) < v < 1 \\ 1 & u \text{ and } v \geq 1 \end{cases}$$

Let $U := F_1(U_{(n)})$ and $V := F_2(U_{(n-1)})$, and define

$$\begin{aligned} F_u(v) &:= P(V \leq v | U = u) \\ &= \frac{\partial}{\partial u} C(u,v) \\ &= \begin{cases} u^{(1-n)/n} \cdot F_2^{-1}(v)^{n-1} & v \leq \phi(u) \\ 1 & v > \phi(u) \end{cases} \end{aligned}$$

to be the conditional distribution of V given $U = u$. Hence the mass of a theoretical sample (u_i, v_i) drawn from C will lie below the curve $\phi(x) = nx^{(n-1)/n} - (n-1)x$.

To generate a sample from this theoretical copula, one can calculate

$$\begin{aligned} F_u^{-1}(w) &= F_2(w^{1/(n-1)}) \\ &= nwu^{(n-1)/n} - (n-1)(w^{n/(n-1)}u). \end{aligned} \tag{6}$$

Then $(U, F_U^{-1}(W)) \sim C$ for $U, W \sim U[0,1]$ independent. Figure 10 shows the convergence behavior of the curve $\phi(x)$ as well as the mass of the simulated theoretical copula points as the sample size n increases. We also prove that $f(x) = x(1 - \ln x)$ is truly the limit of $\phi(x)$ as $n \to \infty$ in lemma 5.3.

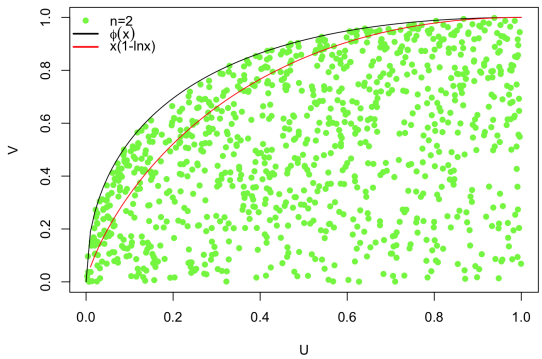

FIGURE 9. Theoretical Copula Generated from $C(u, v)$ when $n = 2$

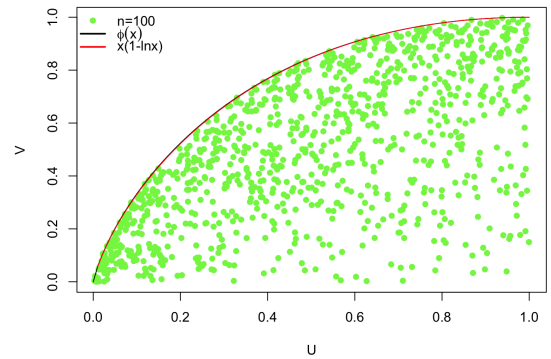

FIGURE 10. Theoretical Copula Generated from $C(u, v)$ when $n = 100$

Lemma 5.3. *Let* $f_n(x) = n(x^{(n-1)/n}) - (n-1)x$. *Then* $f_n(x) \to f(x) = x(1 - \ln(x))$ *uniformly on* $[0, 1]$.

Proof. Observe that

$$\begin{aligned} d_n &= \sup_{x \in [0,1]} |f_n(x) - f(x)| \\ &= \sup_{x \in [0,1]} |nx^{\frac{n-1}{n}} - nx + x\ln(x)| \end{aligned}$$

which attains its max at $x = 1$. Hence $d_n = n - n + \ln(1) = 0 \to 0$ as $n \to \infty$.

□

There is more work to be done to generalize this to the largest r ranked order statistics, $(X_{(n)}, \ldots, X_{(n-r+1)})$. I conjecture that an explicit formula, similar to equation 6, can also be derived to generate theoretical copula points for the largest r ranked order statistics for r larger than 2.

6. Testing This Copula

Now we'd like to get a better understanding of whether the copula $C(u, v)$ found in equation 5 adequately measures the dependence structure of the largest and second largest rainfall measurements in our actual data. Visually from figure 7, we saw that the empirical copula points follow a similar pattern to the theoretically generated copula, potentially suggesting that this copula may be a good fit. However, we can formalize this test by using a goodness-of-fit test based on the Rosenblatt probability transform, as described in Genest et al. [6].

We'd like to test the hypothesis

$$H_0 : (\hat{U}_i, \hat{V}_i) \sim C,$$

where $(\hat{U}_i, \hat{V}_i)$ is the empirical copula of our data, as constructed in section 3.4.

The Rosenblatt transform on a copula pair is defined as $R(U, V) := (E_1, E_2)$ where:

$$\begin{aligned} E_1 &= U \\ E_2 &= \frac{\partial}{\partial u} C(u, v) \,/\, \frac{\partial}{\partial u} C(u, 1) \end{aligned}$$

Observe that, since $u \in (0, 1)$, we have:

$$\begin{aligned} \frac{\partial}{\partial u} C(u, 1) &= \frac{\partial}{\partial u} \int_0^u \mathbb{1}\{1 \geq t(1 - \ln t)\} + \mathbb{1}\{1 < t(1 - \ln t)\} \psi_t^{-1}(1) dt \\ &= \mathbb{1}\{1 \geq u(1 - \ln u)\} + \mathbb{1}\{1 < u(1 - \ln u)\} \psi_u^{-1}(1) \\ &= 1, \end{aligned}$$

so $E_2 = \mathbb{1}\{v \geq u(1 - \ln(u))\} + \mathbb{1}\{v < u(1 - \ln(u))\} \Psi_u^{-1}(v)$.

Due to properties of the Rosenblatt transform, testing H_0 is equivalent to testing

$$R(\hat{U}_i, \hat{V}_i) \sim C_\perp,$$

where $C_\perp$ is the bivariate independence copula [6]. If the copula were a good fit, we would expect the plot of the transformed points to be uniformly distributed on $[0, 1] \times [0, 1]$, which does not appear to be the case as shown in figure 11.

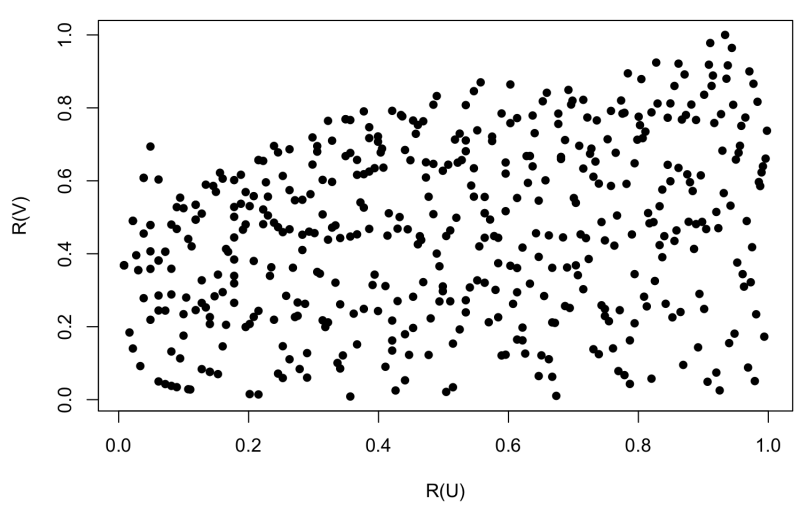

FIGURE 11. Rosenblatt transform applied to the empirical copula from monthly block maxima

To perform this test we use the following test statistic:

$$S_n = \sum_{i=1}^{n} [D_n(R(\hat{u}_i, \hat{v}_i)) - C_\perp(R(\hat{u}_i, \hat{v}_i))]^2,$$

where $D_n(u, v) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{R(\hat{U}_i, \hat{V}_i) \leq (u, v)\}$ is the empirical distribution of the transformed data [6]. Using the bootstrap procedure described in A.4, we find that the p-value is far below 0.05, thus rejecting the hypothesis that this copula is an appropriate model for the underlying dependence structure of our data.

This may be due to other dependencies present in our data, such as time dependence, which were not accounted for in the theoretical calculation of the copula (which was based on independent samples).

7. Introducing Time Dependencies

Another observation from figure 7, is that the empirical copula of our rainfall data lies well below the theoretical limit curve. Compare this to figure 12, in which the empirical copula for an independently sampled Fréchet points lies above the curve $f(x) = x(1 - \ln x)$, or figure 6 where the points sampled from this copula follow the curve quite closely. This may be further evidence that there could be some other underlying dependence structure in our data that is not modeled in the theoretical independent case.

Purely theoretically, we had calculated that the copula (in equation 5) measures the dependence structure of $(X_{(n)}, X_{(n-1)})$, where X_i are sampled *independently.* However, in the case of weather data such as rainfall, we have no guarantee of independence for each measurement. For example, certain weather patterns may affect the rainfall amount on consecutive days, or seasonality changes (e.g. wet season vs. dry season) may mean that measurements made during a rainy season come from a different distribution than those from a dry season.

To see whether time dependency has an effect on the copula model, we can run a few simulations using a moving maximum process: We sample Z_i independently from a standard Fréchet, and then set $X_i = \max(\frac{a}{a+1}Z_{i-1}, \frac{1}{a+1}Z_i)$, creating a max-stable sample of X_i, so that $X_{(n)} = \max(X_1, \ldots, X_n)$ is GEV as $n \to \infty$ [4]. The scalar $a \in R$ controls the time dependency between neighboring X_i ($a = 0$ are independent, while $a = 1$ creates a heavily time dependent sample where the largest and second largest values often occur in consecutive pairs).

In figure 13, we see that as the value of a increases between 0 and 1, the empirical copula falls beneath the curve, perhaps suggesting a similar phenomenon occurring in the empirical copula for the rainfall data. There also appears to be a sweeping pattern as the points draw nearer to the line $y = x$ as $a \to 1$.

However, more work remains to be done to investigate how time dependencies between data points affects the copula model for ranked order statistics.

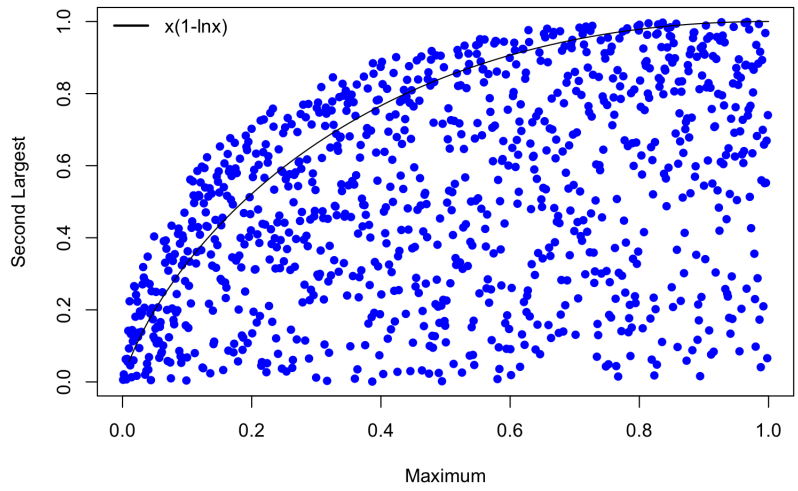

FIGURE 12. Empirical Copula of $(X_{(n)}, X_{(n-1)})$ where X_i are sampled independently from a standard Fréchet.

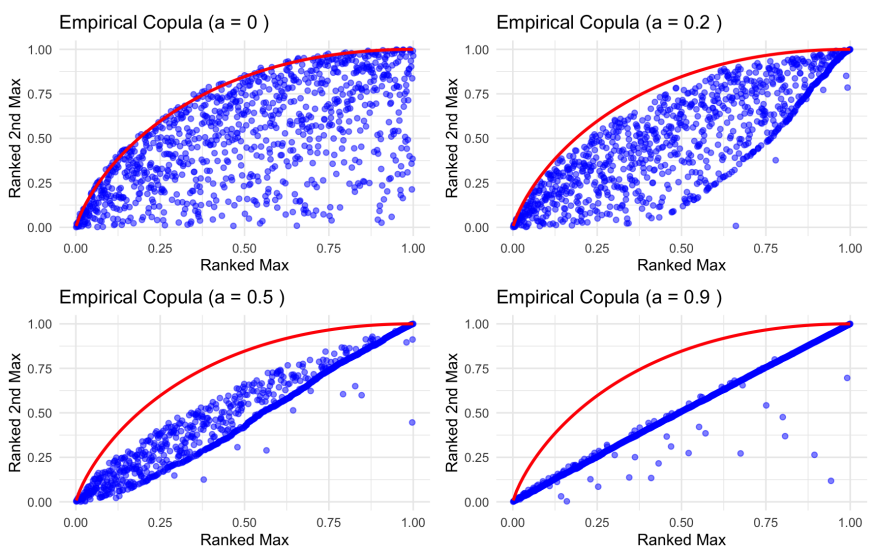

FIGURE 13. Empirical Copula of $(X_{(n)}, X_{(n-1)})$ plotted for a sample of size $n = 1000$ with varying levels of time dependencies.

8. CONCLUSION

In this paper we showed that the copula of the largest and second largest order statistics is non-parametric and follows the curve $f(x) = x(1 - \ln x)$ in the limit. I believe it is possible to use similar techniques as in 4.1 to calculate a non-parametric formulation for the general copula of the largest r ranked order statistics. I am also curious to investigate further how time dependencies affect the structure of this copula, for example the sweeping pattern which appears in figure 13.

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Appendix A. Calculation details

A.1. **Joint Distribution of $X_{(n)}, X_{(n-1)}$.** Let $X_{(n)} = \max(X_1, \ldots, X_n)$ be such that $P((X_{(n)} - b_n)/a_n \leq z) \to G(z)$ as $n \to \infty$, where G is part of the GEV family. Let the joint distribution of $X_{(n)}, X_{(n-1)}$ be

$$F(x, y) = P(\frac{X_{(n)} - b_n}{a_n} \leq x, \frac{X_{(n-1)-b_n}}{a_n} \leq y).$$

(1) If $x = y$:

$$F(x, y) = P(\frac{X_{(n)} - b_n}{a_n} \leq x) \approx G(x).$$

(2) If $y > x$:
You can't have $X_{(n-1)} > X_{(n)}$, so the distribution of $X_{(n-1)}$ is limited by x,

$$F(x, y) = P(\frac{X_{(n)} - b_n}{a_n} \leq x, \frac{X_{(n-1)-b_n}}{a_n} \leq x) \approx G(x).$$

(3) If $y < x$:
Using the Poisson point process described in [4] section 7.3, let $A_u = \{z : z > u\}$ and $N_n(A_u) = \{(\frac{i}{n+1}, \frac{X_i - b_n}{a_n}) : i = 1, \ldots, n\}$ be the sequence of n points which fall in A_u. So under our working assumptions, $N_n(A_u) \to N \sim \text{Poi}(\Lambda(A_u))$ where

$\Lambda(A_u) = (t_2 - t_1)[1 - \xi(\frac{u-\mu}{\sigma})]^{-1/\xi}$. Let $\tau(z) = \Lambda(A_z)$. Note that:

$$\begin{aligned}\Lambda(A_y \backslash A_x) &= \Lambda(A_y) - \Lambda(A_x) \\ &= \tau(y) - \tau(x).\end{aligned}$$

We are interested in:

$$\begin{aligned}F(x,y) &= P(N_n(A_x) = 0, N_n(A_y \backslash A_x) \leq 1) \\ &= P(N_n(A_x) = 0) \cdot P(N_n(A_y \backslash A_x) \leq 1) \\ &= e^{-\tau(x)} \sum_{i=0}^{1} \exp(-\Lambda(A_y \backslash A_x) \frac{\Lambda(A_y \backslash A_x)^i}{i!} \\ &= e^{-\tau(x)}[e^{-\tau(y)+\tau(x)}(1 + \tau(y) - \tau(x))] \\ &= e^{-\tau(y)}[1 + \tau(y) - \tau(x)]\end{aligned}$$

So in conclusion:

$$F(x,y) = \begin{cases} e^{-\tau(x)} & y \geq x \\ e^{-\tau(y)}[1 + \tau(y) - \tau(x)] & y < x \end{cases}$$

A.2. **Partial derivative of $C(u,v)$ with respect to u.**

$$\begin{aligned}C(u,v) &= F(F_X^{-1}(u), F_Y^{-1}(v)) \\ &= \begin{cases} e^{-\tau(F_X^{-1}(u))} & F_Y^{-1}(v) \geq F_X^{-1}(u) \\ e^{-\tau(F_Y^{-1}(v))}(1 + \tau(F_Y^{-1}(v)) - \tau(F_X^{-1}(u))) & F_Y^{-1}(v) < F_X^{-1}(u) \end{cases}\end{aligned}$$

where $\tau(x) = [1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi}$ and $F_X(x) = e^{-\tau(x)}$ and $F_Y(y) = e^{-\tau(y)}[1 + \tau(y)]$. Using the chain rule we have:

$$\frac{\partial}{\partial u}[e^{-\tau(F_X^{-1}(u))}] = e^{-\tau(F_X^{-1}(u))}(-\tau'(F_X^{-1}(u)))[\frac{\partial}{\partial u} F_X^{-1}(u)]. \tag{7}$$

First note that:

$$\begin{aligned}\tau'(x) &= \frac{d}{dx}[1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi} \\ &= \frac{-1}{\xi}[1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi - 1} \cdot \frac{\xi}{\sigma} \\ &= [1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi} \cdot \frac{-1}{\sigma}[1 + \xi(\frac{x-\mu}{\sigma})]^{-1} \\ &= \tau(x) \cdot \frac{-1}{\sigma}[\frac{\sigma}{\sigma + \xi(x-\mu)}] \\ &= \frac{-\tau(x)}{\sigma + \xi(x-\mu)}\end{aligned}$$

And solving for x in $F_X^{-1}(x) = y$ we see that:

$$\begin{aligned}
y &= e^{-\tau(x)} \\
&\Rightarrow \ln y = -[1 + \xi(\frac{x-\mu}{\sigma})]^{-1/\xi} \\
&\Rightarrow (-\ln y)^{-\xi} = 1 + \xi(\frac{x-\mu}{\sigma}) \\
&\Rightarrow x - \mu = \frac{\sigma}{\xi}[-1 + (-\ln y)^{-\xi}] \\
&\Rightarrow x = \mu - \frac{\sigma}{\xi}[1 - (-\ln y)^{-\xi}] := F_X^{-1}(y)
\end{aligned}$$

giving us:

$$\frac{\partial}{\partial u} F_X^{-1}(u) = \frac{\sigma}{u}[-\ln u]^{-\xi-1}.$$

We can also solve:

$$\begin{aligned}
\tau(F_X^{-1}(u)) &= [1 + \xi(\frac{F_X^{-1}(u) - \mu}{\sigma})]^{-1/\xi} \\
&= [1 + \frac{\xi}{\sigma}(\frac{-\sigma}{\xi}(1 - (-\ln u)^{-\xi}))]^{-1/\xi} \\
&= [1 - (1 - (-\ln u)^{-\xi})]^{-1/\xi} \\
&= -\ln u
\end{aligned}$$

Returning to equation 7, we see:

$$\begin{aligned}
\frac{\partial}{\partial u}[e^{-\tau(F_X^{-1}(u))}] &= e^{-\tau(F_X^{-1}(u))} \frac{\tau(F_X^{-1}(u))}{\sigma + \xi(F_X^{-1}(u) - \mu)} \cdot \frac{\sigma}{u}[-\ln u]^{-\xi-1} \\
&= e^{\ln u} \frac{-\ln u}{\sigma + \xi(-\frac{\sigma}{\xi}(1 - (-\ln u)^{-\xi}))} \cdot \frac{\sigma}{u}[-\ln u]^{-\xi-1} \\
&= u \frac{-\ln u}{\sigma - \sigma(1 - (-\ln u)^{-\xi})} \cdot \frac{\sigma}{u}[-\ln u]^{-\xi-1} \\
&= \frac{-\ln u}{\sigma(-\ln u)^{-\xi}} \cdot \sigma(-\ln u)^{1(\xi+1)} \\
&= (-\ln u)^{1+\xi}(-\ln u)^{-(\xi+1)} \\
&= 1
\end{aligned}$$

Next:

$$\begin{aligned}
\frac{\partial}{\partial u}[e^{-\tau(F_Y^{-1}(v))}(1 + \tau(F_Y^{-1}(v)) - \tau(F_X^{-1}(u)))] &= \frac{\partial}{\partial u}[-\tau(F_X^{-1}(u))e^{-\tau(F_Y^{-1}(v))}] \\
&= e^{-\tau(F_Y^{-1}(v))} \frac{\partial}{\partial u}[\ln u] \\
&= \frac{1}{u} e^{-\tau(F_Y^{-1}(v))}
\end{aligned}$$

Then since F_Y and F_X are monotone increasing, we have:

$$\begin{aligned} F_Y^{-1}(v) \geq F_X^{-1}(u) &\iff v \geq F_Y(F_X^{-1}(u)) \\ &\iff v \geq e^{-\tau(F_X^{-1}(u))}[1+\tau(F_X^{-1}(u))] \\ &\iff v \geq e^{\ln u}[1-\ln u] \\ &\iff v \geq u(1-\ln u). \end{aligned}$$

Putting all of this together we see that:

$$\frac{\partial}{\partial u}C(u,v) = \begin{cases} 1 & v \geq u(1-\ln u) \\ \frac{1}{u}e^{-\tau(F_Y^{-1}(v))} & v < u(1-\ln u) \end{cases}$$

A.3. **Inverse of the Conditional Probability.** Let $F_u(v) = \frac{1}{u}e^{-\tau(F_Y^{-1}(v))}$. First we solve for x in $\tau(x) = y$ to find:

$$\begin{aligned} y &= [1+\xi(\frac{x-\mu}{\sigma})]^{-1/\xi} \\ &\Rightarrow \xi(\frac{x-\mu}{\sigma}) = y^{-\xi} - 1 \\ &\Rightarrow \frac{x-\mu}{\sigma} = \frac{y^{-\xi}-1}{\xi} \\ &\Rightarrow \tau^{-1}(y) := \frac{\sigma}{\xi}(y^{-\xi}-1) + \mu \end{aligned}$$

Then we solve for v in $F_u(v) = w$ to find:

$$\begin{aligned} &\frac{1}{u}e^{-\tau(F_Y^{-1}(v))} = w \\ &\Rightarrow -\tau(F_Y^{-1}(v)) = \ln(uw) \\ &\Rightarrow F_Y^{-1}(v) = \tau^{-1}(-\ln(uw)) \\ &\Rightarrow F_Y^{-1}(v) = \frac{\sigma}{\xi}[(-\ln uw)^{-\xi} - 1] + \mu \\ &\Rightarrow v = F_Y(\frac{\sigma}{\xi}[(-\ln uw)^{-\xi} - 1] + \mu) \end{aligned} \tag{8}$$

Observe that:

$$\begin{aligned} \tau(\frac{\sigma}{\xi}[(-\ln uw)^{-\xi} - 1] + \mu) &= [1 + \frac{\xi}{\sigma}(\frac{\sigma}{\xi}[(-\ln uw)^{-\xi} - 1])]^{-1/\xi} \\ &= [1 + ((-\ln uw)^{-\xi} - 1)]^{-1/\xi} \\ &= -\ln uw \end{aligned}$$

Hence returning to equation 8, we see that:

$$\begin{aligned} F_u^{-1}(w) &= F_Y(\frac{\sigma}{\xi}[(-\ln uw)^{-\xi} - 1] + \mu) \\ &= e^{\ln uw}[1 - \ln uw] \\ &= uw(1 - \ln uw). \end{aligned}$$

A.4. **Bootstrapping Procedure.** We want to test $H_0 : R(\hat{U}_i, \hat{V}_i) \sim C_\perp$ i.e. test if $R(\hat{U}_i, \hat{V}_i)$ is uniformly spread on $[0,1]^2$. Under the null hypothesis, the empirical distribution of $R(\hat{U}_i, \hat{V}_i)$ should be "close" to $C_\perp(e_1, e_2) = e_1 e_2$, so we use the following test statistic

$$S_n = \sum_{i=1}^{n} [D_n(R(\hat{u}_i, \hat{v}_i)) - C_\perp(R(\hat{u}_i, \hat{v}_i))]^2,$$

where $D_n(u,v) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{R(\hat{U}_i, \hat{V}_i) \leq (u,v)\}$ is the empirical distribution. We use the following procedure, which can be coded into R, to calculate a p-value for this test statistic:

(1) Calculate $R(\hat{U}_i, \hat{V}_i) := (E_{i1}, E_{i2})$ for $i = 1, \ldots, n$: the Rosenblatt transform of the empirical copula of our actual data.

(2) Calculate S_n. This is the value we want to judge to be too large or not.

(3) Bootstrap for $k = 1, \ldots, K$ (large):
   (3a) Generate (u_i^k, v_i^k) from the hypothesized copula C : a new set of theoretical copula points.

   (3b) Calculate $R(u_i^k, v_i^k) := (E_{i1}^k, E_{i2}^k)$: their Rosenblatt transforms (which in theory should be distributed according to $C_\perp$).
   (3c) Calculate S_n^k of these points.
(4) Then the p-value is the proportion of the theoretical points which are larger than our observed distance:

$$p = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}\{S_n^k > S_n\}.$$

Appendix B. Code

All figures are my own and were produced using R. You can find my code on my GitHub (`https://github.com/helenaheinonen/math470`).